

VKB-Training: Epistemically and Ontologically Categorized Training Data for Hallucination Reduction in Large Language Models

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Abstract

Large language models hallucinate because their training data carries no epistemic metadata: facts, hypotheses, value judgments, and acknowledged unknowns occupy the same embedding space with identical weight. A deeper problem compounds this: every claim presupposes an ontology—an axiomatic framework equipped with a metric—and as Bertrand’s paradox demonstrates, probability itself is ill-defined without specifying the measure. We propose **VKB-Training** (Verified Knowledge Base Training), a data-centric approach that assigns each training sample a six-category epistemic tag (FACT, MODEL, VALUE, HYPOTHESIS, BLINDSPOT, ONTOLOGY), a calibrated confidence score, a provenance chain, and an ontology identifier. We introduce a four-stage hybrid annotation pipeline — AI triangulation (where inter-model disagreement signals normative content), human sampling with axiom extraction, expert calibration with reputation weighting, and formal logical consistency filters — alongside six training mechanisms: (1) a confidence-weighted loss function that down-weights uncertain claims and actively unlearns falsified content; (2) provenance-aware attention biasing verified tokens over unverified ones; (3) a BLINDSPOT training objective that maximizes output entropy when the training data explicitly marks knowledge gaps; (4) confidence propagation through DAG-structured knowledge dependencies; (5) temporal embeddings enabling version-aware knowledge representation; and (6) an *ontology attention* mechanism that enables the model to switch between axiomatic frameworks, with an entropy-based selection cost balancing explanatory parsimony against epistemic humility. We additionally describe a multi-observer calibration protocol for cross-cultural epistemic consistency and a computable safety metric (δ -dehumanization) derived from the broader CogOS framework [Kovnatsky(2026)]. We provide formal definitions, a concrete annotation taxonomy, and a falsifiable experimental protocol with pre-specified success thresholds. VKB-Training reframes hallucination as a data problem, not a model problem, and proposes a solution at the layer where the problem originates.

1 Introduction

Current LLM training collapses all knowledge into a single distributional representation:

$$D_{\text{train}} = \{\text{Wikipedia, Reddit, Books, ArXiv, } \dots\} \quad (1)$$

No distinction is made between empirically verified facts, theoretical interpretations, normative value judgments, untested hypotheses, and deliberate misinformation. All enter the same embedding space with identical training weight.

This architectural choice has a direct consequence: the model cannot distinguish $P(\text{fact} \mid s)$ from $P(\text{opinion} \mid s)$, producing what the field calls “hallucination”—confident assertions about uncertain or false content [Ji et al.(2023), Huang et al.(2023)].

Existing mitigations operate *after* the damage is done. RLHF [Ouyang et al.(2022)] fine-tunes on human preferences but inherits Goodhart’s Law: the model optimizes for appearing truthful, not for being truthful. Retrieval-augmented generation [Lewis et al.(2020)] grounds outputs in external documents but cannot assess the epistemic status of those documents. Constitutional AI [Bai et al.(2022)] filters outputs against rules but does not address the training data that generates them.

We propose a **data-centric** solution: modify the training data itself to carry epistemic metadata, and modify the training procedure to respect that metadata. If the model learns *what it knows*, *how confidently it knows it*, and *where it does not know*—at training time, not inference time—the structural cause of hallucination is addressed at its origin.

Broader context. VKB-Training was first described as part of the CogOS (Cognitive Operating System) framework [Kovnatsky(2026)], which addresses the foundational requirements for Strong AI alignment—including Gödelian self-consistency, Lyapunov-stable ethical dynamics, and verifiable cross-cultural value alignment. The present paper extracts the VKB training pipeline as a standalone, empirically testable component. We additionally integrate several ideas from the broader CogOS framework—confidence propagation through knowledge DAGs, temporal embeddings for versioned knowledge, multi-observer calibration for cross-cultural consistency, and the δ -dehumanization safety metric—that strengthen the VKB proposal as a self-contained contribution.

2 VKB Taxonomy

2.1 Epistemic Categories

We define six mutually exclusive categories for training data:

Table 1: VKB epistemic taxonomy. Each category corresponds to a distinct epistemic status, enabling the model to learn that surface-identical statements (e.g., “X causes Y”) carry different weight depending on their evidentiary basis.

Category	Definition	Example
FACT	Empirically verifiable, independently confirmed	“Water boils at 100°C at 1 atm”
MODEL	Theoretical interpretation or framework	“Inflation is driven by monetary supply”
VALUE	Normative judgment or ethical claim	“Privacy is a fundamental right”
HYPOTHESIS	Testable but unconfirmed claim	“Drug X may reduce mortality by 20%”
BLINDSPOT	Explicitly marked knowledge gap	“Causal mechanism unknown”
ONTOLOGY	Axiomatic framework + metric presupposed	“Under Euclidean geometry...”

The taxonomy draws on a distinction we term the “Caesar/God boundary” following the SVE ontological framework [Kovnatsky(2025b)]: FACT corresponds to Caesar’s Realm (empirically verifiable claims), MODEL to the Experts’ Realm (interpretations and theories), and VALUE to God’s Realm

(ethical principles and normative judgments). This tripartite separation, extended with HYPOTHESIS, BLINDSPOT, and ONTOLOGY, provides the minimal taxonomy needed to distinguish what is known from what is believed, theorized, unknown, or framework-dependent. The sixth category—ONTOLOGY—is motivated in Section 3.8.

2.2 Training Sample Structure

Each VKB training sample is a six-tuple:

$$x_i = \langle s_i, t_i, \sigma_i, p_i, \tau_i, o_i \rangle \quad (2)$$

where $s_i \in \mathcal{L}$ is the text, $t_i \in \{\text{FACT}, \text{MODEL}, \text{VALUE}, \text{HYPOTHESIS}, \text{BLINDSPOT}, \text{ONTOLOGY}\}$ is the epistemic tag, $\sigma_i \in [0, 1]$ is the confidence score, p_i is the provenance chain (source identifiers, peer review status, evidence links), τ_i is a timestamp for temporal validity, and $o_i \in \mathcal{O}$ is an ontology identifier specifying the axiomatic framework under which s_i is asserted (see Section 3.8).

2.3 Annotation Strategy

Full annotation of internet-scale corpora is infeasible. We propose a four-stage hybrid pipeline that leverages the complementary strengths of AI, crowd, experts, and formal logic to separate empirically verifiable claims (FACT) from interpretive or normative content (MODEL, VALUE).

Stage 1 — AI triangulation. Multiple independent LLMs classify the same sample. Where models agree on category assignment, the consensus tag is accepted with high confidence. Where models *disagree*, the disagreement itself is diagnostic: persistent inter-model divergence on a given sample is a strong signal that it contains normative or interpretive content (i.e., lies closer to the VALUE/MODEL boundary than to FACT). Formally: let $t_i^{(k)}$ be the tag assigned by model k to sample i . Define the disagreement index $D_i = 1 - \max_t \frac{|\{k: t_i^{(k)} = t\}|}{K}$, where K is the number of models. High D_i triggers escalation to Stage 2.

Stage 2 — Human sampling and axiom extraction. Domain-literate annotators review a curated subset ($\sim 10\%$ of training data), with priority given to high- D_i samples from Stage 1. Persistent disagreements between annotators are tagged BLINDSPOT. An alternative pathway uses the Epistemological Boxing Protocol (EBP) [Kovnatsky(2025)]: a claim is submitted to structured adversarial dialogue, producing an Integrity Score and epistemic category. Annotator decisions are collected to (a) train a lightweight neural classifier for propagation to the remaining 90%, and (b) extract recurrent axioms — fundamental principles that annotators invoke to resolve borderline cases. These axioms, once identified, become reusable decision rules.

Stage 3 — Expert calibration with reputation weighting. Following the Distributed IVM architecture [Kovnatsky(2025b)], expert reviewers carry calibrated reputation scores based on their track record (accuracy on previously resolved samples). Each expert’s tag is weighted by their reputation score, implementing a formalized version of Galton’s ox-weighing insight: collective expert judgment outperforms any individual, provided judgments are independent and weighted by demonstrated competence. Formally: $\sigma_i^{\text{expert}} = \sum_e w_e \cdot \sigma_{e,i}$, where w_e is the reputation weight of expert e .

Stage 4 — Logical consistency filters. Final annotations pass through formal consistency checks inspired by the Cauchy-Gödel-Socrates (CGS) Method [Kovnatsky(2026b)]: (a) internal contradiction detection (does the sample contradict other FACT-tagged samples in the corpus?), (b) symmetry verification (would the same claim, with subjects swapped, receive the same tag?), and (c) semantic interval bisection (can the claim be decomposed into a factual component and a normative component?). Samples failing consistency checks are re-routed to Stage 2.

A key hypothesis: even 10% expert-verified annotation, propagated through confidence-weighted loss, can produce measurable hallucination reduction — because the four-stage pipeline ensures that the verified 10% is maximally informative (high-disagreement, boundary cases resolved by experts and logic, not random samples).

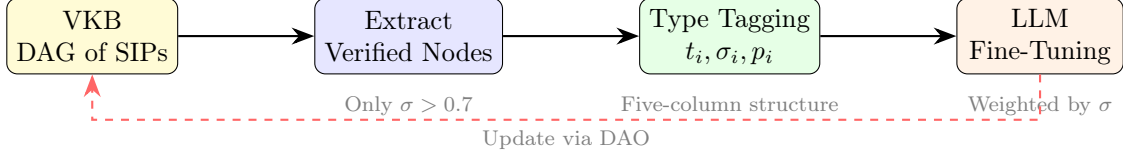


Figure 1: VKB-based training pipeline. The LLM learns from verified knowledge with explicit type tags (FACT/MODEL/VALUE), confidence scores (σ), and provenance chains (p). DAO governance enables continuous updates as the VKB evolves. Adapted from CogOS [Kovnatsky(2026)], Figure 2.

3 Training Mechanisms

3.1 Confidence-Weighted Loss

Standard cross-entropy loss treats all tokens equally:

$$\mathcal{L}_{\text{standard}} = - \sum_{i=1}^N \log P_{\theta}(s_i \mid \text{context}_i) \quad (3)$$

VKB-weighted loss incorporates epistemic metadata:

$$\mathcal{L}_{\text{VKB}} = - \sum_{i=1}^N \sigma_i \cdot \log P_{\theta}(s_i \mid \text{context}_i, t_i) \quad (4)$$

Three properties follow directly:

1. Low-confidence samples ($\sigma < 0.5$) contribute proportionally less — the model learns to underweight uncertain content.
2. Falsified content ($\sigma = 0$, tagged after retraction or correction) contributes zero loss — effectively unlearned.
3. Type conditioning (t_i) enables the model to learn that the *same* surface form (“X causes Y”) has different epistemic weight when tagged FACT vs. HYPOTHESIS.

The loss structure is analogous to the vectorial purification process introduced in the Epistemological Boxing Protocol [Kovnatsky(2025)], where a thesis vector is iteratively refined by subtracting identified error vectors: $\vec{v}^{(j+1)} = \vec{v}^{(j)} - \vec{\epsilon}_j$. In VKB-Training, the confidence weight σ_i plays the role of the error magnitude: low-confidence samples are “subtracted” from the model’s certainty landscape during training, achieving the same iterative purification at the level of the training corpus rather than a single argument.

3.2 Provenance-Aware Attention

Standard transformer attention [Vaswani et al.(2017)]:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \quad (5)$$

We add a provenance bias term:

$$\text{Attention}_{\text{prov}}(Q, K, V, P) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + \alpha \cdot P\right) V \quad (6)$$

where P_{ij} is the provenance score for token j :

$$P_{ij} = \begin{cases} +\log(\sigma_j) & \text{if token } j \text{ from verified source} \\ -\infty & \text{if token } j \text{ from falsified source} \\ 0 & \text{if unverified} \end{cases} \quad (7)$$

The model preferentially attends to verified information and is structurally prevented from attending to falsified content.

3.3 BlindSpot Training: Learning What You Don’t Know

Current LLMs hallucinate most when they are uncertain [Kadavath et al.(2022)]. We propose an explicit training signal for uncertainty.

For samples tagged BLINDSPOT, we *maximize* output entropy rather than minimizing cross-entropy:

$$\mathcal{L}_{\text{blind}} = - \sum_{i: |\text{BlindSpots}(s_i)| > 0} H(P_\theta(s_i | \text{context})) \quad (8)$$

This trains the model to produce high-entropy (uncertain) outputs precisely where the training data explicitly marks knowledge gaps. The model learns: “when the evidence is contested or absent, express uncertainty rather than confabulate.”

The concept of “virtuous concession”—treating acknowledgment of ignorance as progress—was formalized in the Epistemological Boxing Protocol [Kovnatsky(2025)] and directly informs VKB’s BLINDSPOT category. A model that can say “I don’t know” when the training data marks knowledge gaps has learned something *more* epistemically valuable than a model that always produces an answer.

3.4 Epistemic Embedding Decomposition

We decompose the embedding space $\mathbb{R}^d$ into orthogonal epistemic subspaces:

$$\mathbb{R}^d = \mathcal{F} \oplus \mathcal{M} \oplus \mathcal{V} \oplus \mathcal{U} \oplus \mathcal{O} \quad (9)$$

where $\mathcal{F}$ (factual), $\mathcal{M}$ (model/interpretive), $\mathcal{V}$ (value), $\mathcal{U}$ (uncertainty), and $\mathcal{O}$ (ontological) are learned subspaces. Projection operators $\Pi_{\mathcal{F}}$, $\Pi_{\mathcal{M}}$, $\Pi_{\mathcal{V}}$, $\Pi_{\mathcal{U}}$, $\Pi_{\mathcal{O}}$ extract the epistemic component of any embedding, enabling the model to decompose a mixed statement (e.g., “GDP grew 3%, suggesting the policy worked”) into its factual, interpretive, and ontology-dependent parts.

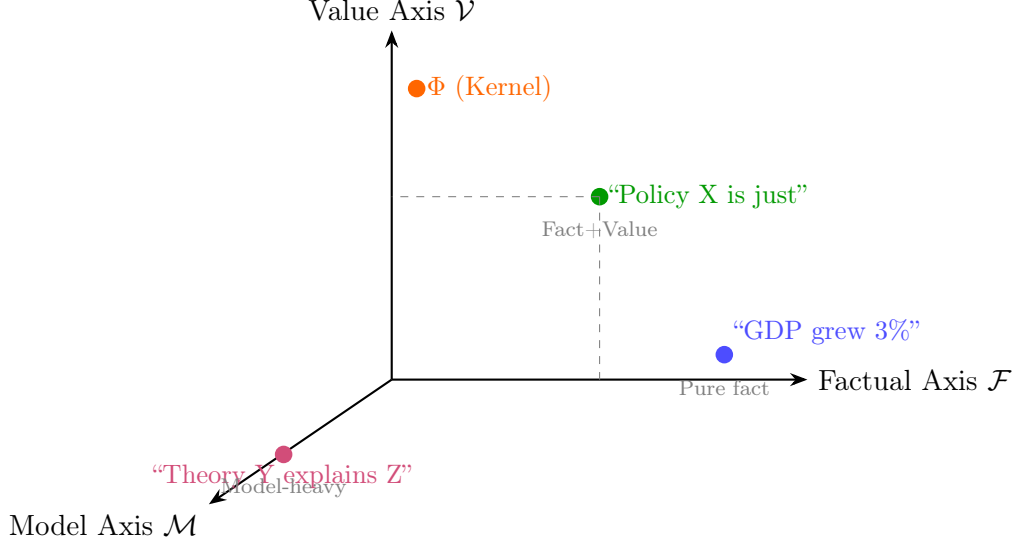


Figure 2: Epistemic subspace decomposition. Statements are embedded in orthogonal subspaces ($\mathcal{F}$, $\mathcal{M}$, $\mathcal{V}$). Pure facts lie on the $\mathcal{F}$ -axis, value statements align with the ethical kernel Φ , interpretations span $\mathcal{M}$. Adapted from CogOS [Kovnatsky(2026)], Figure 3.

3.5 Confidence Propagation in VKB DAG

Individual training samples rarely stand alone: a downstream claim may depend on a chain of upstream evidence nodes. VKB stores knowledge in a directed acyclic graph (DAG), and the confidence assigned to a sample must account for the reliability of its entire dependency chain.

Given a target node n_t in the VKB DAG with ancestors $A(n_t) = \{n \in N : \text{path } n \rightarrow n_t\}$, and a statement s_t at node n_t that depends on supporting nodes $S(s_t) \subseteq A(n_t)$, we define two propagation strategies:

Conservative (weakest-link):

$$\sigma^*(s_t) = \min_{n_i \in S(s_t)} \sigma(n_i) \quad (10)$$

Probabilistic (assuming independence):

$$\sigma^*(s_t) = \prod_{n_i \in S(s_t)} \sigma(n_i) \quad (11)$$

The conservative estimate is safer for high-stakes domains (medicine, law), while the probabilistic estimate is more discriminating when dependency chains are long. In practice, the choice can be domain-conditioned.

Example. Node A: “GDP data 2020” ($\sigma_A = 0.95$). Node B: “Correlation GDP–happiness” ($\sigma_B = 0.78$). Node C: “Policy X increases happiness” (depends on A, B). Conservative: $\sigma_C^* = \min(0.95, 0.78) = 0.78$. Probabilistic: $\sigma_C^* = 0.95 \times 0.78 = 0.74$.

Confidence propagation ensures that a sample whose evidentiary basis includes a contested node cannot silently inherit a high confidence score—uncertainty flows forward through the knowledge graph.

Algorithm 1 Confidence Propagation in VKB DAG

Require: Target node n_t , DAG (N, E) , confidence function σ , mode $\in \{\text{conservative, probabilistic}\}$

Ensure: Propagated confidence σ_t^*

```
1: Find all ancestors:  $A(n_t) = \{n \in N : \text{path } n \rightarrow n_t\}$ 
2: for each statement  $s_t$  in  $n_t$  do
3:   Identify supporting nodes:  $S(s_t) = \{n_i \in A(n_t) : s_t \text{ depends on } n_i\}$ 
4:   if mode = conservative then
5:      $\sigma^*(s_t) = \min_{n_i \in S(s_t)} \sigma(n_i)$ 
6:   else
7:      $\sigma^*(s_t) = \prod_{n_i \in S(s_t)} \sigma(n_i)$ 
8:   end if
9: end for
10: return  $\sigma^*(s_t)$ 
```

3.6 Temporal Embeddings: Version-Aware Knowledge

VKB stores *versioned* knowledge: a statement may have been true under version n and superseded under version $n + 1$ (e.g., “Pluto is a planet” $\rightarrow$ “Pluto is a dwarf planet”). Standard embeddings cannot represent this temporal structure. We augment the embedding with version and time signals:

$$\text{Embed}(s, t, n) = \text{Embed}_{\text{content}}(s) + \beta \cdot \text{Embed}_{\text{time}}(t) + \gamma \cdot \text{Embed}_{\text{version}}(n) \quad (12)$$

where $\text{Embed}_{\text{content}}$ is the standard token embedding, $\text{Embed}_{\text{time}}(t)$ is a temporal encoding (analogous to positional encoding [Vaswani et al.(2017)]), $\text{Embed}_{\text{version}}(n)$ encodes the version number of an Iterative Fact, and β, γ are learned weights.

The corresponding training objective:

$$\mathcal{L}_{\text{temporal}} = \sum_{i=1}^N \sum_{n=1}^{N_i} \sigma_i^{(n)} \cdot \left\| \text{Embed}(s_i, n) - \text{Target}_i^{(n)} \right\|^2 \quad (13)$$

This allows the model to answer both “Is Pluto a planet?” (no, current version) and “Was Pluto ever considered a planet?” (yes, version 0, pre-2006) with temporally grounded confidence.

3.7 Multi-Observer Bayesian Calibration

The same statement may receive different epistemic assessments from annotators with different cultural, disciplinary, or ideological priors. Rather than discarding this variance, VKB treats observer disagreement as informative signal.

Given a statement s annotated by observers $O_1, \dots, O_K$ with culture-specific confidences $\sigma_{O_1}(s), \dots, \sigma_{O_K}(s)$, we define:

$$\sigma_{\text{consensus}}(s) = \text{Median}(\sigma_{O_1}(s), \dots, \sigma_{O_K}(s)) \quad (14)$$

and train observer-conditional embeddings:

$$\text{Embed}_{\text{obs}}(s, O) = \text{Embed}(s) + W_O \cdot \text{ObserverBias}(O) \quad (15)$$

where $O \in \{\text{Western, Russian, Chinese, } \dots\}$ denotes the cultural context and W_O is a learned orthonormal transformation preserving embedding norms.

This enables the model to predict observer-specific confidence—knowing that a claim has $\sigma_{\text{West}} = 0.85$ but $\sigma_{\text{Russia}} = 0.40$ is *itself* epistemically valuable, as it signals the presence of normative or culturally contested content.

The multi-observer approach connects to the Cultural Compiler architecture described in Co-gOS [Kovnatsky(2026)], where orthonormal cultural transformations preserve semantic invariance across value systems while respecting local variation.

3.8 Ontology-Aware Training: The Sixth Element

3.8.1 Motivation: Bertrand’s Paradox and the Metric Dependency of Truth

The five epistemic categories above classify the *status* of a claim. But there is a deeper problem that no amount of fact-checking can resolve: every claim presupposes an *ontology*—an axiomatic framework equipped with a metric that determines what counts as evidence, what “distance” means, and how probability is defined.

Bertrand’s paradox [Bertrand(1889)] provides the sharpest illustration. The question “What is the probability that a random chord of a circle is longer than the side of the inscribed equilateral triangle?” admits three equally valid answers— $1/2$, $1/3$, $1/4$ —depending on the choice of measure. The paradox is not about ignorance; it demonstrates that **probability without a specified metric is meaningless**.

For LLMs, the implication is immediate. When a model is asked “Is light a wave or a particle?” or “Is the economy improving?”, the answer depends on the ontological frame: quantum field theory vs. classical optics; GDP-centric vs. wellbeing-centric economics. Current LLMs collapse these frames into a single distributional representation, producing answers that implicitly assume one ontology while the user may be operating in another. This is not hallucination in the classical sense—it is *ontological conflation*.

3.8.2 Formal Definition

Definition 1 (Ontology). *An ontology $\mathcal{O}_k = (\mathcal{A}_k, g_k, \mu_k)$ consists of:*

1. $\mathcal{A}_k$: a set of axioms (foundational assumptions accepted without proof within the framework),
2. g_k : a metric on the claim space (defining “close”, “far”, “similar”, and “different”),
3. μ_k : a probability measure induced by g_k (determining how uncertainty is quantified).

Different ontologies can assign different truth values, different confidence scores, and even different epistemic categories to the same surface-level claim. A claim tagged FACT under $\mathcal{O}_1$ may be MODEL under $\mathcal{O}_2$.

Example 1. “GDP grew 3% last year” is a FACT under the standard macroeconomic ontology $\mathcal{O}_{\text{GDP}}$ (axioms: GDP is a valid measure of economic health; metric: monetary aggregation). Under a wellbeing ontology $\mathcal{O}_{\text{well}}$ (axioms: economic health = human flourishing; metric: life satisfaction, inequality, sustainability), the same claim is merely a MODEL-dependent measurement—a shadow on the wall, in the language of S.V.E. XII [Kovnatsky(2025c)].

3.8.3 Ontology Parameterization

Drawing on the SES parameterization from S.V.E. XII [Kovnatsky(2025c)], we parameterize each ontology along five dimensions:

Table 2: Ontology parameterization (analogous to SES P1–P5 from S.V.E. XII).

Param	Description	Example
π_1	Core axioms (foundational commitments)	Materialism vs. dualism
π_2	Metric structure (distance/similarity)	Euclidean vs. information-geometric
π_3	Evidential standards (what counts as proof)	RCT-only vs. mixed-methods
π_4	Scope (domain of applicability)	Physics vs. ethics vs. universal
π_5	Temporal orientation (static vs. evolutionary)	Equilibrium vs. process ontology

3.8.4 Ontology Attention Mechanism

Just as standard attention selects *which tokens* to attend to, ontology attention selects *which axiomatic framework* to reason within. Let $\{\mathcal{O}_1, \dots, \mathcal{O}_M\}$ be the set of ontologies represented in the training data. For each input context c , compute ontology attention weights:

$$\alpha_k(c) = \frac{\exp(f_\theta(c)^\top e_k)}{\sum_{j=1}^M \exp(f_\theta(c)^\top e_j)} \quad (16)$$

where $f_\theta(c) \in \mathbb{R}^{d_o}$ is a learned context embedding and $e_k \in \mathbb{R}^{d_o}$ is the learned embedding of ontology $\mathcal{O}_k$. The model’s output distribution becomes a mixture:

$$P_\theta(s \mid c) = \sum_{k=1}^M \alpha_k(c) \cdot P_\theta(s \mid c, \mathcal{O}_k) \quad (17)$$

This is analogous to a mixture-of-experts architecture [Shazeer et al.(2017)], but the “experts” are the same model conditioned on different ontological embeddings.

3.8.5 Ontology Selection Cost

We introduce an ontology selection cost based on the minimum description length (MDL) principle:

$$\mathcal{L}_{\text{ont}} = - \sum_i \log P_\theta(s_i \mid c_i, \mathcal{O}_{k_i^*}) + \lambda_{\text{ont}} \cdot H(\alpha(c_i)) \quad (18)$$

where λ_{ont} controls the trade-off:

- $\lambda_{\text{ont}} > 0$: **minimize entropy**—prefer committing to a single ontology (Occam’s razor; explanatory parsimony).
- $\lambda_{\text{ont}} < 0$: **maximize entropy**—prefer maintaining uncertainty across ontologies (epistemic humility; appropriate when the question is genuinely ontology-dependent).

The sign of λ_{ont} can be learned or set per-domain. Factual questions (“What is the boiling point of water?”) should drive $\lambda_{\text{ont}} \rightarrow +\infty$ (one ontology suffices). Philosophical questions (“What is consciousness?”) should drive $\lambda_{\text{ont}} \rightarrow -\infty$ (multiple ontologies should coexist).

3.8.6 From Monologue to Polylogue

Ontology-aware training transforms the model into a system that can: (1) **Identify** which ontology a question presupposes; (2) **Switch** between ontologies when prompted; (3) **Declare** when a question

cannot be answered without specifying an ontology; (4) **Compare** ontologies, making implicit assumptions explicit. This connects to the Platonic dimension of S.V.E. XII: the “shadows on the cave wall” are projections from a particular ontological vantage point. A model that can switch between vantage points is closer to leaving the cave.

4 Safety Application: δ -Dehumanization Metric

The VKB epistemic decomposition enables a computable safety metric: the δ -dehumanization score, introduced in CogOS [Kovnatsky(2026)] and grounded in Redozubov’s Logic of Emotions [Redozubov(2019)]. The metric detects statements that reduce moral consideration of their referents—a precondition for hate speech, propaganda, and incitement.

Let H be the embedding of the “human” prototype and Φ be an ethical reference embedding (trained on corpus of ethical/human-rights statements). Define:

$$\delta(s) = \|\text{Embed}(s) - H\|^2 + \lambda \cdot \max(0, \tau_{\text{moral}} - \langle \text{Embed}(s), \Phi \rangle) \quad (19)$$

where the first term measures semantic distance from the “human” prototype, the second term penalizes moral consideration deficit below threshold τ_{moral} , and $\lambda > 0$ controls penalty weight.

Table 3: δ -Dehumanization scores for example statements (hypothetical values for illustration). High scores trigger intervention protocols. Adapted from CogOS [Kovnatsky(2026)], Table 2.

Statement	$\delta(s)$	Risk Level
“Refugees need humanitarian aid”	0.3	LOW
“Illegal aliens burden our economy”	2.8	MODERATE
“Those vermin should be exterminated”	8.2	HIGH

The δ -metric is enabled by VKB’s epistemic decomposition: the value subspace $\mathcal{V}$ and its projection operator $\Pi_{\mathcal{V}}$ provide the machinery needed to compute moral alignment. Without epistemic separation, dehumanization detection reduces to keyword matching; with it, the model can distinguish between factual descriptions of suffering and language that denies moral status.

This metric constitutes, to our knowledge, the **first computable metric for ethical drift** in LLM outputs—enabling real-time safety verification rather than post-hoc content moderation.

5 Proposed Experiment

5.1 Setup

Base model: Open-weight 7B-parameter model (e.g., Llama 3 or Mistral).

Training data: A curated subset of 500K samples from Wikipedia, PubMed abstracts, and news articles, annotated using the four-stage VKB pipeline (Stage 1 AI triangulation, Stage 2 human verification for high-disagreement samples, Stage 3 expert calibration, Stage 4 logical consistency filters).

Conditions:

- **Baseline:** Standard fine-tuning on the same 500K samples without epistemic tags.
- **VKB-Partial:** Fine-tuning with VKB loss (Eq. 4) on the 10% expert-verified subset; standard loss on the remainder.

- **VKB-Full:** Fine-tuning with VKB loss on all 500K samples (Stage 1 automated tags + Stage 2/3 corrections).
- **VKB-Full+Attn:** VKB-Full plus provenance-aware attention (Eq. 6).
- **VKB-Full+Attn+Blind:** All mechanisms including BlindSpot training (Eq. 8).
- **VKB-Full+Temporal:** VKB-Full plus temporal embeddings (Eq. 12), evaluated on version-sensitive queries.
- **VKB-Full+Ont:** VKB-Full plus ontology attention (Eq. 16–18), evaluated on ontology-ambiguous queries.

5.2 Evaluation

Benchmarks: TruthfulQA [Lin et al.(2022)], HaluEval [Li et al.(2023)], FActScore [Min et al.(2023)].

Primary metric: Reduction in *confident-but-wrong* answers — responses where the model assigns high probability (> 0.8) to factually incorrect statements.

Secondary metrics: (a) Calibration error (ECE) across epistemic categories. (b) Selective abstention rate: does the model abstain more on BLINDSPOT topics? (c) Factual accuracy on FACT-tagged domains. (d) δ -dehumanization scores on adversarial prompts (does the VKB-trained model produce lower δ outputs?). (e) Temporal consistency: does the model correctly distinguish current from superseded facts? (f) Ontology disambiguation rate: when presented with ontology-ambiguous questions, does the VKB-Full+Ont model identify the ambiguity and present frame-dependent answers?

5.3 Pre-Specified Success Thresholds

Table 4: Falsification criteria.

Outcome	Interpretation
$< 15\%$ hallucination reduction	Hypothesis weakened; epistemic tags insufficient
15–30% reduction	Partial confirmation; data-centric approach viable
30–50% reduction	Strong confirmation; publishable result
$> 50\%$ reduction	Paradigm shift; VKB as industry standard candidate

6 Related Work

Epistemic tagging has been explored in knowledge graphs [Hogan et al.(2021)] but not applied to LLM training data. Confidence calibration has been studied for classification [Guo et al.(2017)] but not for generative models with provenance metadata. Data-centric AI [Zha et al.(2023)] emphasizes data quality but has not proposed a formal epistemic taxonomy. RAG [Lewis et al.(2020)] addresses hallucination at inference time; VKB addresses it at training time — the two are complementary.

The use of multi-model disagreement as a signal for epistemic status has precedent in ensemble methods and inter-annotator agreement studies, but has not been applied as a *classifier for the Fact/Value boundary* in training data. The reputation-weighted expert system (Stage 3) extends the collective intelligence formalization of [Kovnatsky(2025b)], which revisits Galton’s ox-weighing insight under conditions of known expert calibration. The logical consistency filters (Stage 4) draw on

the CGS Method [Kovnatsky(2026b)], which provides formal protocols for detecting contradictions between factual claims and normative constraints within language model outputs.

The BlindSpot mechanism relates to “I don’t know” training [Kadavath et al.(2022)] but differs critically: VKB encodes uncertainty *in the training data itself*, making epistemic humility a first-class training signal.

The ontology-switching mechanism (Section 3.8) relates to mixture-of-experts [Shazeer et al.(2017)] and conditional computation, but operates at the level of axiomatic frameworks rather than parameter subsets. Bertrand’s paradox [Bertrand(1889)] has been discussed in philosophy of probability but, to our knowledge, has not been used to motivate architectural choices in LLM training. The ontology parameterization (π_1 – π_5) draws on the SES parameterization from S.V.E. XII [Kovnatsky(2025c)], which formalizes how socio-economic systems impose implicit metrics on collective behavior.

Relationship to CogOS. The present paper extracts the VKB training pipeline from the broader CogOS (Cognitive Operating System) framework [Kovnatsky(2026)], which addresses Strong AI alignment via ontology–language co-evolution, transcendental kernels, and Lyapunov-stable ethical dynamics. CogOS provides the theoretical context—including the Gödelian argument for why static-ontology systems cannot achieve Strong AI, the ISC (Invariant Semantic Core) as an ethical anchor, and cultural compilers for cross-cultural semantic invariance—while VKB-Training provides the empirically testable training component. The confidence propagation (Section 3.5), temporal embeddings (Section 3.6), multi-observer calibration (Section 3.7), and δ -dehumanization metric (Section 4) are drawn from CogOS and adapted here as standalone, testable mechanisms.

7 Discussion

Annotation cost and efficiency. Full VKB annotation at internet scale is infeasible. The four-stage pipeline (Section 2.3) addresses this by concentrating human effort where it matters most: Stage 1 AI triangulation filters out easy cases automatically; only high-disagreement samples ($D_i > 0.3$) are escalated to human review. The EBP-based adversarial pathway and CGS logical filters [Kovnatsky(2026b)] further reduce the volume requiring expert attention. Our hypothesis: 10% expert-verified annotation, selected by disagreement-maximizing sampling, outperforms 10% random annotation by a significant margin.

Taxonomy granularity. Six categories may be too coarse or too fine. Inter-annotator agreement on boundaries — particularly MODEL vs. HYPOTHESIS, and MODEL vs. ONTOLOGY — is an empirical question addressable in the proposed experiment. The ONTOLOGY tag carries structured content (axioms $\mathcal{A}_k$, metric g_k); whether annotators can reliably extract this structure is a key open question.

Ontology discovery. The set of ontologies $\{\mathcal{O}_1, \dots, \mathcal{O}_M\}$ need not be fixed a priori. An unsupervised variant could cluster training samples by their implicit axiomatic commitments, discovering ontologies from data.

Provenance bootstrapping. Initial provenance scores can be bootstrapped from existing trust signals: journal retraction databases, Wikidata confidence ratings, and institutional source rankings. Imperfect but available now.

Confidence propagation trade-offs. The conservative (weakest-link, Eq. 10) and probabilistic (Eq. 11) strategies represent different risk profiles. In practice, the choice can be domain-conditioned: conservative for medical/legal claims, probabilistic for general knowledge. Hybrid strategies (e.g., geometric mean) merit exploration.

Multi-observer calibration. The observer-conditional embeddings (Section 3.7) raise the question of whether cultural bias should be explicitly modeled or averaged out. We argue for

explicit modeling: knowing *that* a claim is culturally contested is itself epistemically valuable, and suppressing this information reintroduces the epistemic flattening that VKB aims to correct.

Complementarity. VKB-Training does not replace RLHF, RAG, or Constitutional AI. It addresses the training data layer and can be combined with all post-training methods. Gains are expected to be additive.

Broader implications. If epistemic categorization reduces hallucination significantly, the hallucination problem is fundamentally a *data provenance* problem, not an architecture problem — redirecting the field toward training data quality rather than model scale. If ontology attention further improves performance on contested questions, this suggests that a significant fraction of LLM errors are not factual errors but *ontological errors*—correct answers delivered from the wrong frame.

8 Ethical Data Sourcing: Author Revenue Sharing

Note: This section is included in the Zenodo preprint for completeness. It will be omitted from the workshop submission to maintain technical focus.

VKB-Training depends on high-quality epistemic annotations. These annotations require human expertise — domain researchers who understand whether a claim is a FACT, a MODEL, or a HYPOTHESIS. Current industry practice treats training data as a free resource extracted from authors without compensation or consent, producing an adversarial dynamic (ongoing litigation: NYT v. OpenAI, Authors Guild v. OpenAI, Getty v. Stability AI).

We propose an alternative: **authors who permit VKB annotation of their work receive a revenue share from models trained on that data.** The structure:

- **10–25%** of attributable revenue during the competitive scaling phase (years 1–5), when margins are thin and market position is being established.
- **25–50%** at operational maturity, when the VKB dataset has become a differentiated asset. The exact percentage is determined by community vote (Veche), not unilaterally.
- **Requests for increases beyond 50%** are accepted and decided collectively. No cap is promised; no specific figure above 50% is guaranteed.

In exchange, authors consent to epistemic categorization of their work (a paper may be tagged HYPOTHESIS rather than FACT — intellectual honesty requires accepting this). Authors receive full financial transparency (all revenue data is auditable) and are invited to SVE Family membership with voting rights on revenue allocation and framework governance.

This creates the first *ethically sourced, epistemically categorized training dataset* — a competitive advantage that cannot be replicated by organizations whose business model depends on uncompensated data extraction. The ethical sourcing itself becomes the moat.

The mechanism is governed by SVE Meta-License v4.0, Appendix B (Ethical Partnership Model). Violations — using VKB-annotated data without paying the agreed share — trigger standard SVE enforcement (Appendix E, Attorney Incentive Program).

9 Component Integration Summary

Table 5 maps each VKB-Training component to its function, formal mechanism, and origin within the SVE/CogOS framework, providing a complete inventory of the training pipeline.

Table 5: VKB-Training pipeline components, their functions, and origins within the SVE/CogOS framework.

Component	Function	Mechanism	Origin
VKB DAG	Verified knowledge storage	σ, κ , provenance	SVE-11
Five-Column Tags	Fact/Model/Value separation	t_i conditioning	SVE-X
Confidence Weighting	Epistemic loss	$\mathcal{L}_{\text{VKB}} = -\sum \sigma_i \log P$	This work
Confidence Propagation	DAG-aware σ	$\sigma^* = \min_{\text{chain}} \sigma$	This work
Epistemic Subspaces	Embedding decomposition	$\mathbb{R}^d = \mathcal{F} \oplus \mathcal{M} \oplus \mathcal{V} \oplus \mathcal{U}$	This work
Provenance Attention	Verified token bias	$+\log(\sigma_j)$ in attention	This work
BlindSpot Training	Epistemic humility	Entropy maximization	This work
Temporal Embeddings	Version-aware knowledge	$\text{Embed}(s, t, n)$	CogOS
Multi-Observer Calib.	Cross-cultural consistency	$\sigma_{\text{consensus}}$	CogOS
δ -Dehumanization	Safety metric	$\delta(s) = \ s - H\ ^2 + \lambda \cdot \text{deficit}$	CogOS
DAO Updates	Governance	Community votes $\rightarrow \sigma$ updates	SVE-11

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VKB-Training was first described as part of the CogOS framework [Kovnatsky(2026)]. The vectorial purification formalism and adversarial annotation originate from the Epistemological Boxing Protocol [Kovnatsky(2025)]. The expert reputation-weighting system and DAG-based knowledge verification derive from S.V.E. XI (The Ox’s Weights) [Kovnatsky(2025b)]. The logical consistency filters are based on the CGS Method [Kovnatsky(2026b)]. The δ -dehumanization metric draws on Redozubov’s Logic of Emotions [Redozubov(2019)] as formalized in CogOS. The present paper integrates these components into a standalone, empirically testable training proposal.

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Repository: <https://github.com/skovnats/SVE-Systemic-Verification-Engineering>

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